

LARSEN ICE SHELF, Antarctica

WMO: 892620

Lat: 67.013S

Lon: 61.470W

Elev: 148

StdP: 14.62

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -43.5	(c) -39.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 37.4	(k) -1.4	(l) 32.9	(m) 0.0	(n) 3.3	(o) 250	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 11.6	(c) 41.3	(d) N/A	(e) 38.3	(f) N/A	(g) 36.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.1	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	23.8	20.5	-48.1	49.0	4.4	3.6	-51.3	51.5	-53.9	53.6	-56.3	55.6	-59.6	58.2
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	29.6	23.1	10.0	-4.9	-7.0	-13.6	-13.9	-10.5	-2.9	7.4	18.9	27.4
	DBStd	19.12	3.83	6.98	10.22	12.10	14.04	11.54	14.04	14.92	13.97	11.08	7.97	5.52
	HDD50	16341	633	753	1239	1646	1766	1907	1982	1876	1586	1320	933	700
	HDD65	21816	1098	1173	1704	2096	2231	2357	2447	2341	2036	1785	1383	1165
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	8.2	7.6	8.1	8.1	7.7	7.0	7.9	8.6	8.9	8.6	9.1	8.4	8.2
	PrecAvg	27.1	2.1	2.2	2.6	2.2	2.1	1.9	2.0	2.1	2.5	2.9	2.6	1.9
	PrecMax	40.0	4.5	4.3	3.8	4.8	5.1	3.5	4.3	3.7	4.0	5.3	4.7	4.3
(15) Precipitation	PrecMin	17.1	0.5	1.3	1.4	0.7	1.2	0.8	0.6	0.6	1.4	1.8	0.9	0.6
	PrecStd	7.2	1.0	0.7	0.6	0.9	0.8	0.7	0.8	0.7	0.7	0.9	1.0	0.9
	DB	47.2	41.2	37.6	35.4	35.9	32.5	31.4	29.8	35.0	38.1	41.2	45.9	
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	41.8	37.2	33.7	27.1	30.4	18.6	26.6	24.8	30.8	34.5	37.5	40.6	
	2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% DB	37.8	34.8	29.5	18.4	23.7	9.7	18.6	20.1	25.7	30.5	34.7	37.4	
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10% DB	35.7	32.5	26.0	11.6	15.0	2.7	8.9	14.1	19.1	26.2	31.2	35.2	
	10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	MDBR	11.6	12.6	16.1	15.7	17.2	16.8	17.7	19.7	19.2	19.7	17.3	13.5	
	5% DB	16.4	15.7	20.1	25.0	23.9	27.4	29.1	26.4	22.9	23.2	21.7	18.7	
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub	0.270	0.261	0.239	0.194	0.133	N/A	0.131	0.189	0.227	0.245	0.256	0.267	
	taud	2.279	2.291	2.298	2.181	1.914	N/A	1.901	2.120	2.197	2.187	2.209	2.233	
	Ebn at Noon	313	297	271	214	90	0	81	209	269	299	316	318	
	Edh at Noon	36	31	23	13	3	0	3	14	25	34	38	39	
(44) All-Sky Solar Radiation	RadAvg	2455	1707	914	309	44	0	16	180	681	1471	2302	2748	
	RadStd	75	48	26	14	2	0	1	10	22	32	66	82	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page